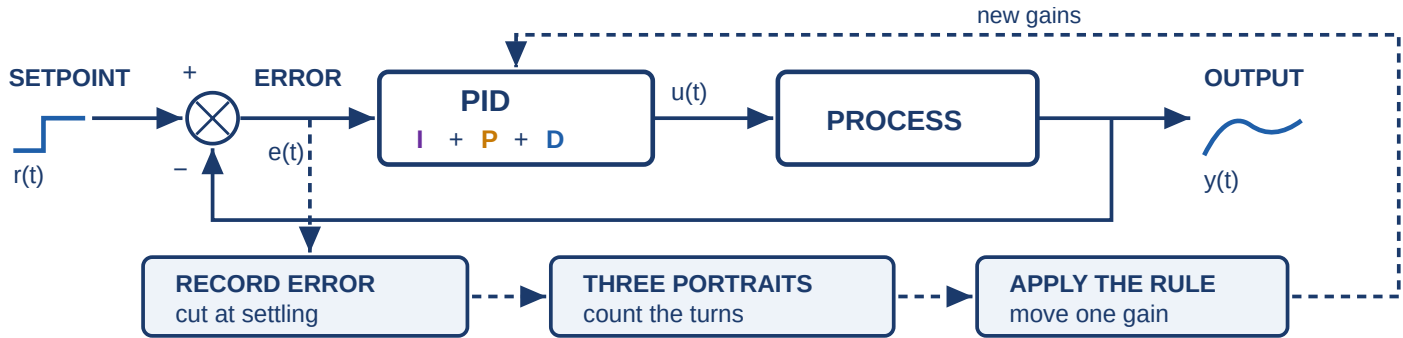


SPIN TUNING

STEP-RESPONSE PHASE-PORTRAIT INSPECTION

Read one closed-loop step response, move one gain, repeat. No model, no relay test, no optimizer.



$$u(t) = K_i \int e(\tau) d\tau + K_p e(t) + K_d de(t)/dt$$

each gain multiplies one signal — and each portrait plots that same signal; derivative filtered, $T_f = K_d/\nu$, $\nu = 10$

K_i Integral $\rightarrow \Gamma_0$

K_p Proportional $\rightarrow \Gamma_1$

K_d Derivative $\rightarrow \Gamma_2$

WHAT EACH PORTRAIT SHOWS

K_i K_p K_d = band (outer arc) red = over limit, cut green = within limit, raise

PORTRAIT	STEP RESPONSE	PACHNER PLOTS — OVER / WITHIN	READING
Γ_0 K_i band $\int e$ vs e		 $N_0 = 2.41$ $N_0 = 0.12$	✓ Slow cycling — the loop hunts around setpoint ✓ Limit $\tilde{N}_0 = 0.50 = 2$ quarter turns ✓ Over limit: cut K_i ↓, raise bands below ↑
Γ_1 K_p band e vs Δe	 $(K_i, K_p, K_d) = (0.35, 0.05, 0.5)$ $(K_i, K_p, K_d) = (0.2, 0.02, 0.7)$	 $N_1 = 2.54$ $N_1 = 0.1$	✓ Ringing — decaying oscillation after the step ✓ Limit $\tilde{N}_1 = 0.75 = 3$ quarter turns ✓ Over limit: cut K_p ↓, raise bands below ↑
Γ_2 K_d band Δe vs $\Delta^2 e$	Both portraits come from these two runs.	 $N_2 = 2.58$ $N_2 = 0.29$	✓ Buzzing — fast chatter on the transient ✓ Limit $\tilde{N}_2 = 1.00 = 4$ quarter turns ✓ Over limit: cut K_d ↓, raise bands below ↑

TUNING GUIDE (SPIN ITERATION)

1

Step it

Close the loop with any gains, step the setpoint, record e

2

Trim it

Screen for runaway, cut where $|e|$ last leaves 2% of peak

3

Count it

Draw the three portraits, count turns
 N_0 N_1 N_2

4

Move it

Use the table below to move K_i K_p K_d , then repeat

RULE — FIRST MATCHING ROW WINS	K_i	K_p	K_d	READING
record grows	↘ ÷2	↘ ÷4	↘ ÷8	runaway — back off, rate hardest
Γ_0 loops	↓	•	•	slow cycling — less reset
Γ_1 loops	↑	↓	•	ringing — less gain
Γ_2 loops	↑	↑	↓	buzzing — less rate
none loops	↑	↑	↑	all quiet — tighten

TIPS

- ✓ Cut only the **lowest** objecting band — higher ones may be leakage
- ✓ Bands below the fault were just measured safe, so raise them
- ✓ Counts carry no units: the same limits fit every loop
- ✓ Any start works: the screen backs the gains off until the record settles
- ✓ No stopping test: stop at any iteration

↑ × y ↓ ÷ y ↓ back off ÷2 ÷4 ÷8 · hold | $y = 1/(1-\beta)$, $\beta = 0.1$; $\tilde{N} = (0.5, 0.75, 1.0)$, $\varepsilon = 0.1$, $\delta = 0.02$, $\nu = 10$

Pachner, Ota, Dostál, Havlena — "Model-Free PID Tuning by Step-Response Inspection," submitted to *Journal of Process Control*, 2026.

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